

Build Bright University

Mobile Programming - Exam Answers

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Subject: Mobile Programming

Part 1: Multiple Choice Questions

Question 1

Android Application Components មានដូចជា (Android Application Components include)

Answer: A, B, C, D - Activity, Service, Content Provider, Broadcast Receiver

The 4 main Android Application Components are: Activity, Service, Content Provider, and Broadcast Receiver. **Fragment is NOT a component** - it's a modular section of an activity.

Question 2

Android Life Cycle មានដូចជា (Android Life Cycle methods include)

Answer: A, B, C, D, E, F, G - onCreate(), onStart(), onResume(), onPause(), onStop(), onRestart(), onDestroy()

The 7 Activity lifecycle methods are: onCreate, onStart, onResume, onPause, onStop, onRestart, onDestroy. **onFragmentRestart() and onActivityRestart() do NOT exist** - these are fake methods.

Question 3

Android Layout មានដូចជា (Android Layout types include)

Answer: A, B, C, D, E - Linear Layout, Relative Layout, Frame Layout, Table Layout, Grid Layout

Valid Android layouts: LinearLayout, RelativeLayout, FrameLayout, TableLayout, GridLayout (also ConstraintLayout). **"Activity Layout" and "List View Layout" are NOT real layout types.**

Question 4

Intent ចែកចេញជា ២ប្រភេទ (Intent is divided into 2 types)

Answer: A, B - Explicit Intent, Implicit Intent

Explicit Intent: Specifies the exact component (class) to start.

Implicit Intent: Declares a general action, letting the system find a suitable component.

"Activity Intent" does NOT exist.

Question 5

សរសេរកូដបើក Login Activity ថ្មី (Write code to open new Login Activity)

Answer: A - `Intent intent = new Intent(MainActivity.this, LoginActivity.class);`
`startActivity(intent);`

startActivity(intent) is used to start a single activity.

startActivities(intent) is used to start multiple activities at once and requires an Intent array, not a single Intent.

Question 6

Adapter ក្នុង Android ប្រើជាមួយ (Adapter in Android is used with)

Answer: A, B, C - List View, RecyclerView, Grid View

Adapters are used with AdapterViews: ListView, RecyclerView, GridView, Spinner, etc. **GridLayout is NOT an AdapterView** - it's a layout container that doesn't use adapters.

Question 7

Orientation នៅក្នុង Linear Layout (Orientation in Linear Layout)

Answer: B, C - horizontal, vertical

LinearLayout's android:orientation attribute accepts only two values: **horizontal** or **vertical**. layout_gravity, gravity, and layout_width are different attributes, not orientation values.

Question 8

Toast នៅក្នុង Android ប្រើសំរាប់ (Toast in Android is used for)

Answer: A - បង្ហាញ Message នៅលើ Screen (Display Message on Screen)

Toast displays a brief message to the user on screen. It's NOT for logging - **Log class** is used for logging in Android.

Question 9

ដើម្បីធ្វើការទំនាក់ទំនងទៅកាន់ Server នៅក្នុង Android យើងប្រើជា (For server communication in Android we use)

Answer: A, C - Retrofit, Http Request

Retrofit is a popular HTTP client library for Android.

HTTP Request (URLConnection, OkHttp) is valid for server communication.

Note: "Android Valley" is likely a typo for **"Volley"** - Google's HTTP library. If meant to be Volley, then A, B, C would all be correct.

Question 10

4 Points

អ្នកបង្កើត Flutter (Flutter creator)

Answer: A - Google

Flutter was created and is maintained by **Google**. It was first released in May 2017.

Question 11

4 Points

ពាក្យពេញនាមសម្រាប់ JDK គឺ (JDK stands for)

Answer: C - Software Development Kit

Error in Question: JDK actually stands for "**Java Development Kit**", not "Software Development Kit".
Option C is the closest/best available answer. SDK = Software Development Kit.

Question 12

កន្លែង Configuration font, image, dependency នៅក្នុង Flutter គឺ (Configuration file for font, image, dependency in Flutter)

Answer: B - pubspec.yaml

pubspec.yaml is the configuration file in Flutter/Dart projects for managing dependencies, assets (fonts, images), and project metadata.

Question 13

ភាសាដែលសរសេរកម្មវិធី Flutter គឺ (Programming language for Flutter)

Answer: B - Dart

Flutter apps are written in **Dart**, a language also created by Google. Flutter itself is NOT a programming language.

Question 14

តើយើងគួរប្រើប្រាស់ Widget មួយសម្រាប់អោយ value ផ្លាស់ប្តូរនៅ screen mobile? (Which Widget for changing values on screen?)

Answer: B - Stateful widget

StatefulWidget is used when UI needs to change dynamically (mutable state).

StatelessWidget is for static UI that doesn't change.

"Statebuild widget" does NOT exist.

Question 15

4 Points

ជ្រើសរើសកូដសំរាប់ Navigator ទៅកាន់ Screen ដទៃ (Select code for Navigator to another Screen)

Answer: A - `Navigator.push(context, MaterialPageRoute(builder: (context)=>OtherScreen()));`

Navigator.push() adds a new route to the navigation stack (go to new screen).

Navigator.pop() removes current route (go back), NOT for navigating forward.

Question 16

ជ្រើសរើសកូដសំរាប់ Navigator មកកាន់ Screen ដើមវិញ (Select code for Navigator back to original Screen)

Answer: B - `Navigator.pop(context);`

Navigator.pop(context) removes the current route and returns to the previous screen.

`Navigator.pop(context, true)` is used when you want to return a value to the previous screen.

Question 17

យើងប្រើប្រាស់ Dependency `easy_localization` សំរាប់ (`easy_localization` dependency is used for)

Answer: B - ប្តូរភាសា (Change language)

easy_localization is a Flutter package for internationalization (i18n) - translating apps into multiple languages.

Question 18

យើងប្រើប្រាស់ Dependency `localstorage` សំរាប់ (`localstorage` dependency is used for)

Answer: B - ផ្ទុកទិន្នន័យនៅ Local Storage (Store data in Local Storage)

localstorage package stores data persistently on the device's local storage.

"ផ្ទុកទិន្នន័យបណ្តោះអាសន្ន" (temporary storage) would be for in-memory storage, not localstorage.

Question 19

ដើម្បីសុំ Permission access ទៅកាន់ Internet នៅក្នុង Android គឺ (To request Internet permission in Android)

Answer: A - `<uses-permission android:name="android.permission.INTERNET" />`

INTERNET permission grants network access.

WRITE_EXTERNAL_STORAGE is for writing to external storage, not internet access.

Question 20

ជ្រើសរើសថ្លើយដែលអ្នកគិតថាត្រឹមត្រូវ នៅពេលដែលយើងផ្តល់តម្លៃតាម Constructor នៅក្នុង Flutter (Correct way to pass values via Constructor in Flutter)

Answer: A - `Slide slide = Slide(name: "Slide1", imageUrl: "url1");`

The constructor uses **named parameters** with curly braces: `Slide({this.name, this.imageUrl})`

Named parameters MUST be called with their names: `name: "value"`

Option B uses positional syntax which won't work with named parameters.

Part 2: Coding Question

Question 21

ចូរបង្កើត *Teacher Class Java* ដែលមាន *Data Member (Id, Name, Phone, Email)*, *Constructor (Default Constructor, Constructor all parameter)*, *Method (Getter, Setter)*

```
public class Teacher {  
    // Data Members  
    private int id;  
    private String name;  
    private String phone;  
    private String email;  
  
    // Default Constructor  
    public Teacher() {  
    }  
  
    // Constructor with all parameters  
    public Teacher(int id, String name, String phone, String email) {  
        this.id = id;  
        this.name = name;  
        this.phone = phone;  
        this.email = email;  
    }  
  
    // Getters  
    public int getId() {  
        return id;  
    }  
  
    public String getName() {  
        return name;  
    }  
  
    public String getPhone() {  
        return phone;  
    }  
  
    public String getEmail() {  
        return email;  
    }  
  
    // Setters  
    public void setId(int id) {  
        this.id = id;  
    }  
}
```

```
public void setName(String name) {  
    this.name = name;  
}  
  
public void setPhone(String phone) {  
    this.phone = phone;  
}  
  
public void setEmail(String email) {  
    this.email = email;  
}  
}
```


Question 21 (Alternate Version)

ចូរបង្កើត *Student Class* Dart ដែលមាន *Data Member (Id, Name, Status)*, *Constructor (Default Constructor, Constructor all parameter)*, *Method (Getter, Setter)*

```
class Student {
  // Data Members
  int? _id;
  String? _name;
  String? _status;

  // Default Constructor
  Student();

  // Constructor with all parameters
  Student.allParams(int id, String name, String status) {
    _id = id;
    _name = name;
    _status = status;
  }

  // Getters
  int? get id => _id;
  String? get name => _name;
  String? get status => _status;

  // Setters
  set id(int? value) => _id = value;
  set name(String? value) => _name = value;
  set status(String? value) => _status = value;
}
```

Quick Answer Summary

Q#	Answer	Q#	Answer
1	A, B, C, D	11	C
2	A, B, C, D, E, F, G	12	B
3	A, B, C, D, E	13	B

Q#	Answer	Q#	Answer
4	A, B	14	B
5	A	15	A
6	A, B, C	16	B
7	B, C	17	B
8	A	18	B
9	A, C	19	A
10	A	20	A

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